Block Sparse Attention

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1 Analysis

We evaluate three configurations on the standard suite and report the average accuracy across five random seeds.

The estimator that we analyse in this section is

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} w_i z_i, \tag{1}$$

and the weights are chosen so that the variance is minimised over the simplex.

We now state the main guarantee of the paper.

For every step size that satisfies the stated bound the iterates satisfy

$$\|x_t - x^\star\|^2 \le (1 - \eta \mu)^t \|x_0 - x^\star\|^2, \tag{2}$$

and the constant does not depend on the dimension of the problem.

Under these assumptions the analysis of the previous section applies verbatim. The kernel is implemented once and reused for both training and inference in all of the experiments that we report.

Training with the sparse kernel from scratch converges in the same number of steps as the dense baseline, which we verified on three seeds.

Model	Acc	F1
Dense	81.2	79.4
Sparse	83.7	81.9

Table 1: Accuracy and F1 on the held-out split.